

5-minute script — SpectralQ

9 slides shown · 16 skipped · 56 keypresses · 4:50 with 10 sec spare

→ reveals the next item on the current slide. **Shift + →** jumps to the next slide, ignoring whatever reveals are left.

Skipping is what **Shift + →** is for. Without it, moving from slide 07 to slide 13 costs 25 presses because plain → walks every reveal of every slide in between.

Shown: 00 · 04 · 06 · 07 · 13 · 16 · 17 · 20 · 24 **Skipped:** 01 02 03 · 05 · 08 09 10 11 12 · 14 15 · 18 19 · 21 22 23

00 · Title — 15 sec · no reveals

We're SpectralQ. One question: does replacing a layer of a physics neural network with a quantum circuit actually help? We ran it as a controlled ablation — two fluid equations, three seeds each. Short answer is no, and we can explain why.

→→ **Press Shift + → four times** (past 01, 02, 03) to land on **04**.

04 · What a QAPINN changes — 25 sec · 4 reveals

[1] A PINN's first layer is a learned matrix multiply. [2] We replace that one layer with a variational quantum circuit. [3] Everything downstream is byte-identical — same width, depth, optimiser, seeds, training budget. [4] So this is an ablation, not a benchmark. Any difference has exactly one cause.

→→ **Press Shift + → twice** (past 05) to land on **06**.

06 · The five experiments — 30 sec · 4 reveals

[1] Two PDEs at opposite ends of one axis. Heat is smooth — exactly two spatial modes. Burgers forms a shock, so it's broadband. [2] E1 and E2 ask whether the quantum layer helps on each. [3] E3 is the control — is the effect actually quantum, or would periodic activations do the same thing? [4] E4 and E5 test whether our bandwidth formula predicts behaviour at all.

Three seeds each, one machine.

→→ **Press Shift + → once** to land on **07**.

07 · The answer, up front — 30 sec · 3 reveals

Conclusion first, then I'll earn it.

[1] One number — the bandwidth K — tells you before training what the circuit can represent. [2] It does not predict an advantage. There wasn't one. Classical won or tied in every configuration. [3] At roughly twenty times the compute.

Say "there wasn't one" flatly. No wince.

→→ **Press Shift + → six times** (past 08–12) to land on **13**.

13 · The mechanism — 45 sec · 4 reveals

Our bandwidth sweep — and a flaw we found in it.

[1] The ladder changes qubit count alongside K, so the big step at K=4 is also a two-to-four qubit step. We can't attribute that to bandwidth. [2] So hold qubits fixed at four and vary K alone. [3] K=2 gives 0.024. K=4 gives 0.012. K=8 gives 0.016. [4] Minimum exactly at K=4 — the highest mode in the target — then the rise at K=8 that capacity theory predicts.

That's the mechanism, uncontaminated.

→→ **Press Shift + → three times** (past 14, 15) to land on **16**.

16 · The result we almost reported — 45 sec · 4 reveals

The one that wins you the room. Don't rush it.

This is what we actually learned.

[1] We had a 7.8 percent quantum win on Burgers. It was in our draft. [2] It was an artifact of stopping L-BFGS at 500 iterations — training was still descending. [3] That vertical drop is the L-BFGS phase. The classical model gains most from it because it had the most headroom left, so the early cutoff penalised the model that was going to win. [4] We tested it, classical error dropped twelve-fold, and the win vanished.

Most teams would have shipped the win.

Pause after that line.

→→ **Press Shift + → once** to land on **17**.

17 · The scoreboard — 35 sec · 5 reveals

[1] Heat is a statistical tie — not a win in either direction. [2] Burgers, classical wins 3.1 times, t equals 4.8.

[3] Cost, twenty times, on identical hardware. [4] Quantum uses 27 percent fewer parameters. [5]

Memory and out-of-domain generalisation we did not measure, so we don't claim them.

→→ **Press Shift + → three times** (past 18, 19) to land on **20**.

20 · The honest answer — 40 sec · 4 reveals

The thesis.

[1] The quantum layer replaces a free learned map with a fixed band-limited basis. That is a restriction. [2] A restriction only pays off if it fits your problem better than what it replaced. Target inside K, we get parity. [3] Outside it, we lose. [4] And if a band-limited prior is genuinely what you want, SIREN gives you one for a twentieth of the cost.

There's no regime in our results where band-limiting wins.

→→ **Press Shift + → four times** (past 21, 22, 23) to land on **24**.

24 · Conclusion — 25 sec · 4 reveals

[1] A quantum layer is a constraint you have to earn. Here, it wasn't. [2] But the formula predicts what the circuit can represent, and we verified that directly. [3] It correctly predicts where quantum comes closest. [4] And we caught a confound that would have handed us a false positive.

That's the result. Thank you.

Stop. Don't add "any questions."

Press sheet

Slide	Talk	Then
00	—	Shift+→ ×4
04	→ ×4	Shift+→ ×2
06	→ ×4	Shift+→ ×1
07	→ ×3	Shift+→ ×6
13	→ ×4	Shift+→ ×3
16	→ ×4	Shift+→ ×1
17	→ ×5	Shift+→ ×3
20	→ ×4	Shift+→ ×4
24	→ ×4	end

32 → · 24 Shift+→ · 56 total.

What was cut, and why it's safe

Cut	Why it survives being cut
01, 02 — motivation, research question	Judges already know why CFD matters. Slide 00 carries the question.
03 — what is a PINN	Only needed for a lay audience. Cut for a technical panel; if the room is mixed, restore it and drop slide 06 instead.
05 — architecture deep-dive	Slide 04 states the change. Detail is in the report.
08, 09 — Fourier theorem, K formula	Slide 13 shows the formula working. The theory is Schuld et al., not our contribution.
10 — measured spectrum	Strong evidence, but slide 13 already carries the mechanism. First slide to restore if you get 7 minutes.
11, 12 — heat testbed, sweep method	Slide 13 restates what it needs.
14, 15 — Burgers testbed and result	Slide 17 has the $3.1\times$ number.
18 — reproducibility (CV 24–82%)	Genuinely novel, but it's a secondary finding. Second to restore.
19 — explainability probes	Both probes are indirect; weakest evidence in the deck.
21, 22, 23 — recommendation, limitations, future work	Limitations belong in Q&A, where they land better as answers.

Running long? Drop slide 06 (saves 30 sec and 5 presses). The story holds without it. **Running short?** Restore 10, then 18.

Full-length script: *PRESENTER-NOTES.md*